



TRAINING PROGRAMME

Linux Automation with Ansible (EX294)

Automate Red Hat Enterprise Linux administration with Ansible, from a Git repository

42 HOURS

FOR RHEL administrators who want to automate their daily work and prepare EX294

PREPARES FOR Red Hat Certified Advanced System Administrator in Ansible (EX294)



Mohamed Aziz BEN HAH

mohamedazizbenhaha.netlify.app

Overview

This programme teaches participants to automate the administration of Red Hat Enterprise Linux with Ansible, following the EX294 exam objectives. They set up a control node with ansible-navigator, execution environments, VS Code with an Ansible development container and Git, then write inventories, playbooks, variables and templates, protect secrets with Ansible Vault, organise their code in roles and content collections, and use the RHEL system roles. Every chapter ends with a lab on a small fleet of RHEL 10 machines. A timed mock EX294 exam measures readiness, and the programme closes with a final project: a three-tier environment automated from a Git repository and proved idempotent.

At a glance

#	PHASE	FOCUS	HOURS
01	Ansible Foundations	Control node, tooling, Git, inventories and ad hoc commands	6 h
02	Writing Playbooks	Plays, handlers, variables, facts and task control	9 h
03	Templates, Secrets & Roles	Jinja2 templates, Ansible Vault, roles and content collections	6 h
04	Administration Tasks & Exam Practice	RHEL system roles, troubleshooting, automated administration, mock exam	9 h
05	Final Project	A three-tier environment automated from Git, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		42 h

Key facts

AUDIENCE	RHEL administrators who want to automate their daily work and prepare EX294
PREREQUISITES	RHCSA level (course: RHCSA Part 2: Red Hat System Administration II, or RHCSA Rapid Track): the administration tasks themselves are not taught again, only their automation
DURATION	42 h: 36 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs; one timed mock exam
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM minimum, hardware virtualization, Podman or Docker for the VS Code development container) with a RHEL 10 control node and three managed nodes with spare disks, registered with each participant's no-cost Red Hat Developer Subscription for Individuals; Git server provided by the trainer; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase, a timed mock EX294 exam and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can automate RHEL administration with idempotent playbooks, roles, collections and Vault kept in Git, and sit the EX294 exam

PHASE 01

6 H

Ansible Foundations

Participants set up the tools they will use for the whole course: a control node with ansible-navigator and execution environments, VS Code with an Ansible development container, and a Git repository for their automation code. They then describe the lab fleet in inventories and run their first ad hoc commands. Validated lab: an inventory with groups and variables, committed to Git, that reaches every managed node with privilege escalation.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Ansible Architecture & Tooling	Control node, managed nodes, ansible.cfg, ansible-navigator and ansible-navigator.yml, execution environments, VS Code with an Ansible development container, Git basics (clone, add, commit, push)	Lecture + Lab	3 h
1.2	Inventories	Static INI and YAML inventories, groups, host and group variables, dynamic inventory	Lecture + Lab	1.5 h
1.3	Managed Nodes & Ad Hoc Commands	Distributing SSH keys, privilege escalation, modules, ansible-doc and finding modules in collections	Lecture + Lab	1.5 h

Writing Playbooks

Participants write playbooks that describe the desired state of a server and can be run again safely. They use variables and facts to adapt to each host, then control execution with loops, conditions, blocks and error handling. Validated lab: a playbook that installs and configures a web server on the web group, restarts it only when its configuration changes, and reports no change on a second run.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Playbooks	Plays, tasks, handlers, idempotency, check mode, running with ansible-navigator and ansible-playbook	Lecture + Lab	3 h
2.2	Variables & Facts	Variable precedence, registered variables, facts, magic variables	Lecture + Lab	3 h
2.3	Task Control	Loops, conditionals, blocks and rescue, error handling, tags	Lecture + Lab	3 h

Templates, Secrets & Roles

Participants generate configuration files from Jinja2 templates, keep passwords out of the repository with Ansible Vault, and reorganise their playbooks into roles that use content collections. Validated lab: the web server playbook refactored into a role with a templated configuration and a Vault-encrypted password, installed with its collections from a requirements file.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Templates	Jinja2, filters, lookups	Lecture + Lab	1.5 h
3.2	Ansible Vault	Encrypting variables and files, vault IDs	Lecture + Lab	1.5 h
3.3	Roles & Content Collections	Role structure, creating roles, Ansible Galaxy, installing collections, requirements files	Lecture + Lab	3 h

PHASE 04

9 H

Administration Tasks & Exam Practice

Participants automate the administration tasks they already know by hand: packages, services, firewall, storage, users, scheduled jobs and SELinux, using the RHEL system roles where they exist. They learn to debug failing playbooks, then sit a timed mock EX294 exam. Validated lab: a playbook that applies a complete baseline (time, SELinux, firewall, users, storage) to every managed node.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	RHEL System Roles	timesync, selinux, network and storage roles	Lecture + Lab	1.5 h
4.2	Troubleshooting Playbooks	Debugging, verbosity, check and diff modes	Lecture + Lab	1.5 h
4.3	Automating Administration	Packages and repositories, services, firewall rules, filesystems and storage, file content, archiving, scheduled tasks, SELinux, users and groups	Lecture + Lab	3 h
4.4	Mock Exam	Timed EX294 mock exam and debrief	Lab	3 h

PHASE 05

6 H

Final Project

Each participant automates a three-tier environment from their Git repository, reusing the roles and playbooks from the phase labs, then presents it as they would to an operations team.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, design the inventory and roles, then automate the web, application and database tiers with collections, Vault, templates and system roles	Project	3 h
5.2	Prove & Document	Clean run on fresh nodes, a second run with no changes, README and role documentation	Project	1.5 h
5.3	Demo & Review	Live run from the Git repository; questions and feedback	Project	1.5 h

FINAL PROJECT

A Three-Tier Environment as Code

Web, application and database servers configured entirely from a Git repository with roles, content collections, Vault-protected secrets, templates and RHEL system roles; a second run reports no changes.

STACK RHEL 10, Ansible, ansible-navigator, execution environments, Ansible Vault, RHEL system roles, Jinja2, Git, VS Code

DELIVERABLE Git repository with playbooks and roles, README, live run